

Software Deployment Tools

Adversaries may gain access to and use centralized software suites installed within an enterprise to execute commands and move laterally through the network. Configuration management and software deployment applications may be used in an enterprise network or cloud environment for routine administration purposes. These systems may also be integrated into CI/CD pipelines. Examples of such solutions include: SCCM, HBSS, Altiris, AWS Systems Manager, Microsoft Intune, Azure Arc, and GCP Deployment Manager.

Access to network-wide or enterprise-wide endpoint management software may enable an adversary to achieve remote code execution on all connected systems. The access may be used to laterally move to other systems, gather information, or cause a specific effect, such as wiping the hard drives on all endpoints.

SaaS-based configuration management services may allow for broad [Cloud Administration Command](#) on cloud-hosted instances, as well as the execution of arbitrary commands on on-premises endpoints. For example, Microsoft Configuration Manager allows Global or Intune Administrators to run scripts as SYSTEM on on-premises devices joined to Entra ID.^[1] Such services may also utilize [Web Protocols](#) to communicate back to adversary owned infrastructure.^[2]

Network infrastructure devices may also have configuration management tools that can be similarly abused by adversaries.^[3]

The permissions required for this action vary by system configuration; local credentials may be sufficient with direct access to the third-party system, or specific domain credentials may be required. However, the system may require an administrative account to log in or to access specific functionality.

Procedure Examples

ID	Name	Description
G0050	APT32	APT32 compromised McAfee ePO to move laterally by distributing malware as a software deployment task. ^[4]
C0018	C0018	During C0018 , the threat actors used PDQ Deploy to move AvosLocker and tools across the network. ^[5]
G1051	Medusa Group	Medusa Group has utilized software deployment and management solutions to deploy their encryption payload to include BigFix and PDQ Deploy. ^[6]
G0129	Mustang Panda	Mustang Panda has leveraged legitimate software tools such as AntiVirus Agents, Security Services, and App Development tools to execute scripts and to side-load DLLs. ^{[7][8]}
G0034	Sandworm Team	Sandworm Team has used the commercially available tool RemoteExec for agentless remote code execution. ^[9]
G1057	ShinyHunters	ShinyHunters has abused software deployment tools for lateral movement. ^[10]
G0091	Silence	Silence has used RAdmin, a remote software tool used to remotely control workstations and ATMs. ^[11]
G0028	Threat Group-1314	Threat Group-1314 actors used a victim's endpoint management platform, Altiris, for lateral movement. ^[12]

ID: T1072

Sub-techniques: No sub-techniques

① Tactics: [Execution](#), [Lateral Movement](#)

① Platforms: Linux, Network Devices, SaaS, Windows, macOS

Contributors: Joe Gumke, U.S. Bank; Shane Tully, @securitygypsy; Tamir Yehuda

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Mitigations

ID	Mitigation	Description
M1015	Active Directory Configuration	Ensure proper system and access isolation for critical network systems through use of group policy.
M1033	Limit Software Installation	Restrict the use of third-party software suites installed within an enterprise network.
M1032	Multi-factor Authentication	Ensure proper system and access isolation for critical network systems through use of multi-factor authentication.
M1030	Network Segmentation	Ensure proper system isolation for critical network systems through use of firewalls.
M1027	Password Policies	Verify that account credentials that may be used to access deployment systems are unique and not used throughout the enterprise network.
M1026	Privileged Account Management	Grant access to application deployment systems only to a limited number of authorized administrators.
M1029	Remote Data Storage	If the application deployment system can be configured to deploy only signed binaries, then ensure that the trusted signing certificates are not co-located with the application deployment system and are instead located on a system that cannot be accessed remotely or to which remote access is tightly controlled.
M1051	Update Software	Patch deployment systems regularly to prevent potential remote access through Exploitation for Privilege Escalation .
M1018	User Account Management	Ensure that any accounts used by third-party providers to access these systems are traceable to the third-party and are not used throughout the network or used by other third-party providers in the same environment. Ensure there are regular reviews of accounts provisioned to these systems to verify continued business need, and ensure there is governance to trace de-provisioning of access that is no longer required. Ensure proper system and access isolation for critical network systems through use of account privilege separation.
M1017	User Training	Have a strict approval policy for use of deployment systems.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0223	Detection of Adversary Abuse of Software Deployment Tools	AN0623	Detects SCCM, Intune, or remote push execution spawning scripts or binaries from SYSTEM context or unusual consoles (e.g., cmtrace.exe launching PowerShell or cmd.exe).
		AN0624	Detects remote scripts or binaries deployed via Puppet, Chef, Ansible, or shell scripts from orchestration servers executing outside maintenance windows or in unmanaged nodes.
		AN0625	Detects script or binary execution initiated via JAMF, Munki, or custom MDM agents outside of baseline, or JAMF launching new Terminal or osascript processes from remote command payloads.
		AN0626	Detects cloud-native software deployment or management (e.g., SSM Run Command, Intune) initiating script execution on endpoints outside expected org IDs, admin groups.

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